

STAT 637 - HW 2.2

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Show that the following distributions belong to the exponential family. For each distribution, identify: a natural parameter, a sufficient statistic, the mean, and the variance using the equations shown in class

Exponential Family: $f(y; \theta) = a(\theta)b(y)\exp\{\sum_{j=1}^k w_j(\theta)t_j(y)\}$

Where $c(\theta) = \ln(a(\theta))$:

Mean: $\mathbb{E}(t(y)) = -\frac{c'(\theta)}{w'(\theta)}$

Var: $Var(t(y)) = \frac{w''(\theta)c'(\theta) - w'(\theta)c''(\theta)}{(w'(\theta))^3}$

(a) Gamma with unknown rate β , fixed α : $f(y; \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-y\beta}$

Exponential Family:

- $a(\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}$
- $b(y) = y^{\alpha-1}$
- $w_1(\beta) = \beta$
- $t_1(y) = -y$

Natural Parameter: $\eta = \beta$

Sufficient statistic: $T(y) = \sum y$

Mean:

$$c'(\beta) = \frac{d}{d\beta} \ln\left(\frac{\beta^\alpha}{\Gamma(\alpha)}\right) = \frac{d}{d\beta} \alpha \ln(\beta) - \ln(\Gamma(\alpha)) = \frac{\alpha}{\beta}$$

$$w'(\beta) = \frac{d}{d\beta} \beta = 1$$

$$\mathbb{E}[y] = -(-1) \frac{\frac{\alpha}{\beta}}{1} = \frac{\alpha}{\beta}$$

Variance:

$$c''(\beta) = \frac{d}{d\beta} \frac{\alpha}{\beta} = -\frac{\alpha}{\beta^2}$$

$$w''(\beta) = \frac{d}{d\beta} 1 = 0$$

$$Var(y) = \frac{0-1(-\frac{\alpha}{\beta^2})}{1^3} = \frac{\alpha}{\beta^2}$$

(b) Poisson: $f(y; \lambda) = \frac{\lambda^y}{y!} e^{-\lambda}$

$$\lambda^y = e^{y \ln(\lambda)} \implies f(y; \lambda) = \frac{1}{y!} e^{-\lambda + y \ln(\lambda)}$$

Exponential Family:

- $a(\lambda) = 1$
- $b(y) = \frac{1}{y!}$
- $w_1(\lambda) = \lambda, w_2(\lambda) = \ln(\lambda)$
- $t_1(y) = -1, t_2(y) = y$

Natural Parameter: $\eta = (\lambda, \ln(\lambda))$

Sufficient statistic: $T(y) = \sum y$

Mean:

$$c'(\lambda) = \frac{d}{d\lambda} \ln(1) = 1$$

$$w'(\lambda) = \frac{d}{d\lambda} \ln(\lambda) = \frac{1}{\lambda}$$

$$\mathbb{E}[y] = \frac{1}{\frac{1}{\lambda}} = \lambda$$

Variance:

$$c''(\lambda) = \frac{d}{d\lambda} 1 = 0$$

$$w''(\lambda) = \frac{d}{d\lambda} \frac{1}{\lambda} = -\frac{1}{\lambda^2}$$

$$Var(y) = \frac{-\frac{1}{\lambda^2} - 0}{(-\frac{1}{\lambda})^3} = \frac{\frac{1}{\lambda^2}}{\frac{1}{\lambda^3}} = \lambda$$

(c) Negative Binomial with fixed r : $f(y; \pi) = \binom{y+r-1}{r-1} \pi^r (1-\pi)^y$

$$(1-\pi)^y = e^{y \ln(1-\pi)} \implies f(y; \pi) = \binom{y+r-1}{r-1} \pi^r e^{y \ln(1-\pi)}$$

Exponential Family:

- $a(\pi) = \pi^r$
- $b(y) = \binom{y+r-1}{r-1}$
- $w_1(\pi) = \ln(1-\pi)$
- $t_1(y) = y$

Natural Parameter: $\eta = \ln(1-\pi)$

Sufficient statistic: $T(y) = \sum y$

Mean:

$$c'(\pi) = \frac{d}{d\pi} \ln(\pi^r) = \frac{d}{d\pi} r \ln(\pi) = \frac{r}{\pi}$$

$$w'(\pi) = \frac{d}{d\pi} \ln(1 - \pi) = -\frac{1}{1 - \pi}$$

$$\mathbb{E}[y] = -\frac{\frac{r}{\pi}}{-\frac{1}{1 - \pi}} = \frac{r(1 - \pi)}{\pi}$$

Variance:

$$c''(\pi) = \frac{d}{d\pi} \frac{r}{\pi} = -\frac{r}{\pi^2}$$

$$w''(\pi) = \frac{d}{d\pi} -\frac{1}{1 - \pi} = \frac{1}{(1 - \pi)^2}$$

$$Var(y) = \frac{\frac{1}{(1 - \pi)^2} \frac{r}{\pi} - \frac{r}{\pi^2} \frac{1}{1 - \pi}}{\left(\frac{1}{1 - \pi}\right)^3} = \frac{r(1 - \pi)^3}{\pi^2(1 - \pi)^2} = \frac{r(1 - \pi)}{\pi^2}$$

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Poisson: $Var(Y) = \mathbb{E}[Y]$

Binomial: $Var(Y) = \mathbb{E}[Y](1 - p)$

Negative Binomial: $Var(Y) = \frac{\mathbb{E}[Y]}{p}$

- (a) These models differ in a ratio of the 'success' parameter. In Poisson, the mean equals the variance, but in Binomial the variance is always smaller than the mean, and Negative Binomial the variance is larger.
- (b) Poisson may be inappropriate in samples where $\mathbb{E}[Y]$ is very large, because this would also result in a very large variance.
- (c) If the variability is larger than the mean, I would look into a Negative Binomial because it has a variance larger than the mean.